

Linux for DevOps

Lesson 3: Package Management & Updates

Package Management & Updates - Study Notes

Key Concepts

1. **Package Managers** – Tools that automate the installation, upgrading, configuration, and removal of software packages on a Linux system.
2. **Repository (Repo)** – A collection of pre-built packages that a package manager can fetch from, usually hosted on the internet or a local network.
3. **Dependencies & Conflicts** – Packages often rely on other packages (dependencies) and may be incompatible with certain versions of other packages (conflicts).
4. **Update vs Upgrade** – `update` refreshes the package database; `upgrade` actually installs newer package versions.
5. **Package Formats** – Different distributions use different package formats (`.deb` for Debian/Ubuntu, `.rpm` for Red Hat/Fedora, etc.) and corresponding managers (`apt`, `yum`, `dnf`).

Summary

Package management is the backbone of maintaining a Linux system. It allows users to add, update, or remove software while automatically handling dependencies and ensuring system stability. Debian-based systems use `apt` (Advanced Package Tool) to work with `.deb` packages, whereas Red Hat-based systems use `yum` (Yellowdog Updater, Modified) or the newer `dnf` (Dandified YUM) to manage `.rpm` packages. Each tool has a similar set of commands—install, update, upgrade, remove—but differs in syntax and some advanced features.

Keeping packages up to date is essential for security and performance. The typical workflow involves refreshing the package list, upgrading installed packages, and optionally removing unused or orphaned packages. Understanding how to use these tools effectively empowers you to maintain a clean, secure, and efficient system.

Important Points

- **Updating the package list**
 - `apt update` – refreshes the Debian/Ubuntu package index.
 - `yum check-update` or `dnf check-update` – lists available updates without installing them.
- **Installing a package**
 - `apt install <pkg>` – installs a package and its dependencies.
 - `yum install <pkg>` / `dnf install <pkg>` – same for RPM-based systems.
- **Upgrading packages**
 - `apt upgrade` – upgrades all upgradable packages.
 - `apt full-upgrade` – handles dependency changes that require removing or installing additional packages.

- ``yum upgrade`` / ``dnf upgrade`` – upgrades all packages; ``dnf`` automatically resolves dependency conflicts.

- **Removing packages**

- ``apt remove <pkg>`` – removes the package but keeps configuration files.

- ``apt purge <pkg>`` – removes the package and its configuration files.

- ``yum remove <pkg>`` / ``dnf remove <pkg>`` – removes the package and its dependencies if no longer needed.

- **Cleaning up**

- ``apt autoremove`` – removes packages that were automatically installed to satisfy dependencies but are no longer needed.

- ``apt clean`` – clears the local repository of retrieved package files.

- ``yum clean all`` / ``dnf clean all`` – similar cleanup for RPM caches.

- **Repository management**

- Debian: ``/etc/apt/sources.list`` and ``/etc/apt/sources.list.d/``.

- RPM: ``/etc/yum.repos.d/`` (yum) or ``/etc/dnf/repos.d/`` (dnf).

- **Security updates** – Regularly run ``apt update && apt upgrade`` or ``dnf update`` to patch vulnerabilities.

Examples

1. Installing a Web Server on Ubuntu

```
```bash
```

#### Update package list

```
sudo apt update
```

#### Install Apache HTTP Server

```
sudo apt install apache2
```

```
```
```

Explanation:

``apt update`` fetches the latest package lists from the configured repositories.

``apt install apache2`` pulls the ``apache2`` package and any required dependencies, then configures it automatically.

2. Updating All Packages on Fedora

```
```bash
```

#### Refresh metadata and upgrade everything

```
sudo dnf upgrade --refresh
```

```
```
```

Explanation:

``dnf upgrade`` updates all installed packages to the newest available versions.

`--refresh` forces a metadata refresh, ensuring the latest package information is used.

Quick Review

1. **What command does `apt update` perform?**
2. **How does `apt full-upgrade` differ from `apt upgrade`?**
3. **Which command removes a package and its configuration files on Debian?**
4. **What is the purpose of `dnf clean all`?**
5. **Name one file that lists repositories for an RPM based system.**

Further Reading

- Debian Package Management: `apt` manual pages (`man apt`, `man apt-get`, `man apt-cache`)
- Red /Hat Package Management: `yum` vs `dnf` comparison (`man yum`, `man dnf`)
- Advanced Repository Configuration (PPA, EPEL, RPM Fusion)
- Package Signing & GPG verification
- Automating updates with `cron` or `systemd` timers
- Containerized package management (Snap, Flatpak, AppImage)

Feel free to dive into these topics to deepen your understanding of Linux package ecosystems!

